

◆ Defining an independent IM section — merge vs. split threshold

Data: Hong Daehwa, 7-day wrist HR (7/13–7/19), ~306K samples

Pipeline: M_phys (RF) → residual → 300s/150s window → 3-state GMM

Result: 85 IM bursts → 84 inter-burst gaps

Gap stats: mean 114.8 min / median 25.0 min / std 226.0 (right-skewed)

Threshold method comparison (log-gap, n=84):

GMM (2-comp): 26.4 min — BIC 305.89 (best fit) → Adopted

KDE: unimodal density, no valley → not usable

IQR: 234.4 min — ~9× GMM, over-conservative

Paper's 300s / 5-min rule (Sec. 3.6.3) = sensor non-wear session split, NOT an IM-merge criterion

Method	Threshold	Statistical basis	Verdict
GMM (log-gap, 2-comp.)	26.4 min	BIC = 305.89 (lowest)	Adopted
KDE	n/a	Unimodal density - no valley	Not usable
IQR (Q3+1.5xIQR)	234.4 min	Outlier rule (not mixture-aware)	Over-conservative (~9x GMM)

◆ Feature significance across Stable / Ambiguous / IM (one-way ANOVA)

Method: 300s window mean/variability → aligned to GMM IM level → ANOVA

All 6 features significant at $p < .005$

Data	Stable	Ambiguous	IM	p
HR (Watch)	Baseline	Gradual rise	Sharp deviation (3x)	$< .005$
EDA mean (Watch)	Baseline conductance	Similar level	Slight increase	$< .005$
Gyroscope variability (Watch)	Near-static	Increasing	Sharp increase (2x)	$< .005$
EDA variability (Watch)	Near-static	Rising (7x)	Sharp rise (22x)	$< .005$
Geomagnetic variability (Phone)	Stable	Rising (7x)	Large shift (26x)	$< .005$
Skin temperature (Watch)	Normal	Mild decrease	Marked decrease (vasoconstriction)	$< .005$

EDA variability ↑ 22× — strongest sympathetic-arousal marker

Skin temp ↓ — only decreasing feature (vasoconstriction)

Gyro / geomagnetic ↑ — possible movement confound

Next: add motion + EDA + skin-temp to M_phys as covariates

Accelerometer (3.3GB) unavailable → EDA used instead

◆ Lab stress-induction protocol — 2 participants, Hanyang YK Intercollege, 7/23

Protocol: TSST (Public Speaking + Mental Arithmetic) + Cold Pressor Test (CPT) + Rest

Bomi Kim (18, iOS): Speech 13:37–13:41 · Math 13:57–14:01 · CPT 14:11–14:15 · Rest 14:15–14:45

Nahyun Kim (19, iOS): Speech 13:32–13:36 · Math 13:52–13:56 · CPT 14:06–14:10 · Rest 14:15–14:45

M_{phys} (baseline HR model) on lab data: Test $R^2 = 0.83$ (Nahyun) / 0.97 (Bomi)

AIES-style uncertainty sampling: 24% of data → 99% of full AUC-ROC (0.889 vs. 0.898)